# Multimodal Deception Detection

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Github: https://github.com/akalra501/toLieOrNotToLie

## 1. Project Description and Motivation

Deception detection is a problem of significant practical importance across law enforcement, border security, legal proceedings, and human resources. Despite the urgency, existing tools remain fundamentally flawed. The polygraph, the most widely deployed instrument, is subjective, physiologically invasive, and legally inadmissible in most jurisdictions. Critically, human accuracy at identifying deceptive behavior averages only 54%, barely above chance, highlighting the need for objective automated systems.

Existing unimodal AI approaches, which analyze either facial expressions or voice alone, are limited by their inability to capture cross-modal patterns that may be more discriminative than any single channel. This project is motivated by the hypothesis that fusing facial, acoustic, and behavioral signals can outperform any individual modality and represent a step toward a practically deployable deception detection framework.

The learning task is formally defined as binary supervised classification: given a video clip of a person making a statement in a courtroom context, predict whether the statement is deceptive (label = 1) or truthful (label = 0). Labels are derived from courtroom case outcomes and corroborating evidence, providing ground truth that is independent of subjective annotation.

## 2. Data and Learning Problem

### 2.1 Datasets

Two datasets are used in this project:

- Real-life Trial dataset: 121 short video clips (61 deceptive, 60 truthful) of courtroom testimony. The dataset is nearly class-balanced. Each clip is labeled based on verified case outcomes.
- Dolos dataset: 1,446 video clips with behavioral annotations, used for cross-dataset generalization experiments. Combined with the Real-life Trial data on 27 shared features, the merged dataset totals 1,567 samples.

The Real-life Trial dataset is the primary dataset for model development. Its small size (121 clips) represents the central challenge of this project, necessitating rigorous cross-validation and careful regularization to prevent overfitting.

### 2.2 Features and Modalities

Features are extracted from three modalities:

- Behavioral annotations: 39 binary features encoding facial expressions, head pose, gaze direction, and hand movement patterns, annotated frame-by-frame.

- Audio signals: Handcrafted features including 20 MFCCs, spectral centroid, rolloff, bandwidth, contrast, zero crossing rate, RMS energy, pitch (F0) statistics, pause ratio and duration, and voice quality measures (jitter, shimmer, HNR) extracted via librosa and Praat/parselmouth.

- Facial features: 713 continuous-valued features extracted from OpenFace (Action Units, head pose, facial landmarks) and DeepFace (embedding representations).

### 2.3 Core Challenges

- Small sample size: 121 clips limits model capacity and generalization. Stratified k-fold cross-validation is used throughout to maximize data utilization.

- Cross-dataset generalization: Behavioral patterns may not transfer across different annotation teams and recording conditions. The Dolos dataset tests this explicitly.

- Temporal dynamics: Binary clip-level labels discard within-clip temporal structure. A truthful person may pause or hesitate within a clip without the clip being deceptive overall.

- Speaker confound: Different speakers appear across truth and deception conditions, meaning models may partially learn speaker identity rather than deception signal.

## 3. Methodology

### 3.1 Behavioral Pipeline

The behavioral pipeline operates on the 39 binary annotation features. A nested stratified 5-fold cross-validation scheme is used on the 121 Real-life Trial samples. Within each outer fold, Recursive Feature Elimination with Cross-Validation (RFECV) is applied to select a stable subset of features, reducing from 39 to 11 features consistently selected across folds. Hyperparameter tuning is performed inside each inner fold to prevent data leakage. An SVM classifier is trained on the selected features.

For cross-dataset generalization, both datasets are aligned on 27 shared features across the combined 1,567 samples. The same nested CV pipeline is applied to test whether behavioral deception patterns generalize across annotation teams and recording conditions. The cross-training condition (SVM trained on Dolos, tested on Real-life Trial) yields substantially lower performance (accuracy 0.58, AUC 0.599), indicating significant domain shift between the two annotation frameworks.

### 3.2 Audio Pipeline

Audio is extracted from MP4 clips as 16kHz mono WAV files using ffmpeg. Two feature sets are computed:

- Handcrafted features (134 total): prosodic features (pitch mean, std, range, voiced ratio, pause count, duration), spectral features (20 MFCCs + deltas, spectral centroid, rolloff, bandwidth, contrast, chroma, ZCR), energy features (RMS statistics, trend), and voice quality features (jitter local/RAP/PPQ5, shimmer local/APQ3/APQ5, HNR mean/std) via parselmouth/Praat.

- Correlated features are removed using variance thresholding (threshold = 0.01) and pairwise Pearson correlation filtering (threshold = 0.90), reducing from 134 to 79 features.

A repeated stratified 5-fold CV with 5 repeats is used, with hyperparameters tuned on inner folds. An SVM with RBF kernel is the primary classifier. The audio pipeline achieves 72.7% accuracy and 0.819 AUC, substantially below the visual pipeline, consistent with the literature suggesting facial cues are more discriminative than acoustic cues for courtroom testimony.

### 3.3 Video Pipeline

The video pipeline processes 713 features from OpenFace (Action Units AU01-AU45, head pose Rx/Ry/Rz, and 68 facial landmarks) and DeepFace (face embedding). A 3-layer Multilayer Perceptron (MLP) is trained with the following architecture:

- Dense(128, ReLU) - Batch Normalization - Dropout(0.4)
- Dense(64, ReLU) - Batch Normalization - Dropout(0.3)
- Dense(32, ReLU) - Dropout(0.2)
- Dense(1, Sigmoid) - binary output

The objective function is binary cross-entropy loss:

$$\min_\theta \ (1/N) \ \Sigma \ L(f_\theta(X_i), Y_i) + R(\theta)$$

where L is binary cross-entropy, f_θ is the MLP, and R(θ) is an L2 weight decay penalty combined with dropout regularization. The Adam optimizer (lr = 0.001) is used. Early stopping with patience = 15 monitors validation loss to prevent overfitting.

Advanced methodology - Batch Normalization: Following Ioffe and Szegedy (2015), Batch Normalization is applied after the first two dense layers. This normalizes activations per mini-batch during training, reducing internal covariate shift, stabilizing the gradient signal, and allowing higher learning rates. This technique was not covered in the core course curriculum and provides empirical training stability improvements over the baseline without batch normalization. Stratified 5-fold CV is used, with StandardScaler fitted on training folds only.

### 3.4 Late Fusion

Late fusion combines predictions from the visual NN and audio SVM by averaging their predicted probability scores. Three fusion strategies are evaluated: simple averaging, weighted averaging (weights tuned on validation folds), and majority voting. The fused system is evaluated on the same 5-fold CV scheme as the unimodal models.

## 4. Results and Analysis

### 4.1 Unimodal Performance

Table 1 summarizes unimodal classifier performance under stratified 5-fold CV on 121 samples.

| Method | Accuracy | F1 Score | AUC | Features |
|---|---|---|---|---|
| NN: Combined (OF+DF) | 0.9174 | 0.9167 | 0.9735 | 713 |
| SVM: Behavioral (non-DOLOS) | 0.818 | 0.811 | 0.880 | 39 |
| SVM: Audio | 0.7273 | 0.7360 | 0.8189 | 134 |
| SVM: Behavioral (Cross training) | 0.58 | 0.696 | 0.599 | 27 |

***Table 1: Unimodal Classifier Performance***

The neural network trained on combined OpenFace and DeepFace features achieves the best performance across all metrics. Behavioral features without cross-dataset transfer perform competitively (82% accuracy), while the audio SVM is the weakest unimodal system. The cross-training condition confirms significant domain shift between the two behavioral annotation frameworks.

## 4.2 Late Fusion Results

Late fusion results are evaluated across three aggregation strategies. Simple and weighted averaging achieve approximately 90% accuracy and 0.90 F1, which is competitive but does not surpass the visual NN baseline (91.7%). Majority voting performs worst at 81% accuracy. The score distribution analysis reveals why fusion does not improve over the visual baseline: the Visual NN produces near-perfect class separation (truthful predictions cluster near 0, deceptive near 1), while the Audio SVM predictions are heavily overlapping around 0.4-0.6. Fusing a high-confidence discriminative signal with an uncertain one degrades rather than improves decision boundaries.

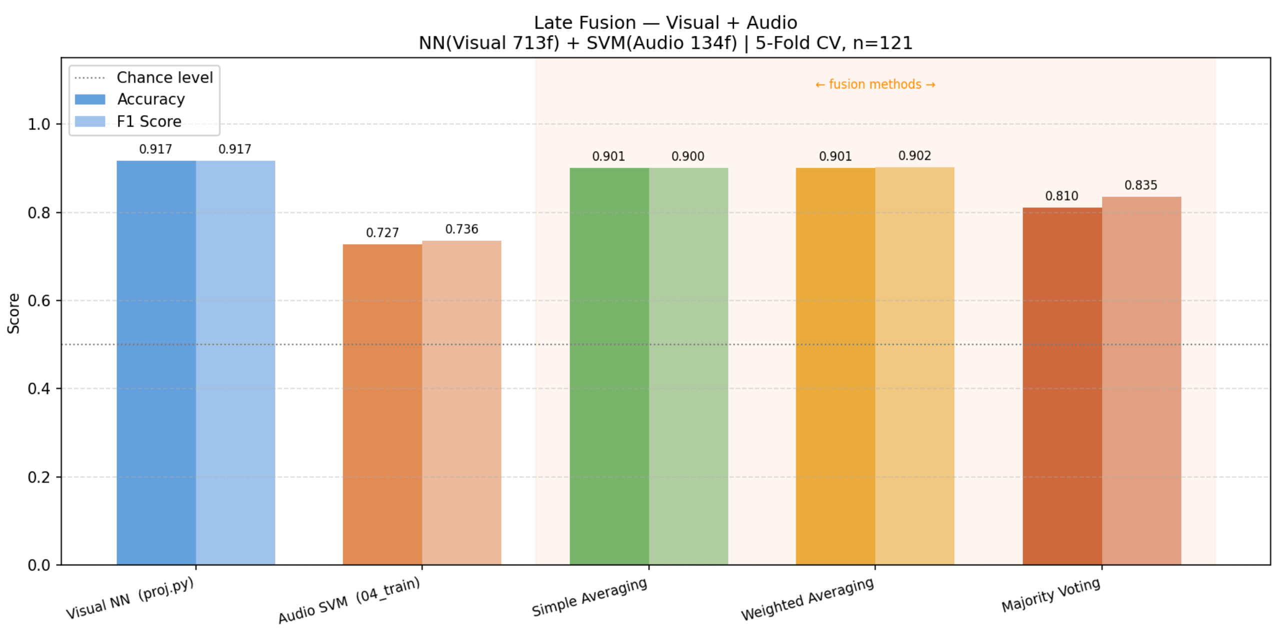


***Figure 1: Late Fusion vs Unimodel results***

Figure 3 presents the predicted probability distributions for each modality and the fused model, showing how confidently each classifier separates truthful from deceptive samples. For the Visual Neural Network, the distributions are almost entirely bimodal, truthful samples cluster sharply at P(deceptive) ≈ 0, while deceptive samples concentrate at P(deceptive) ≈ 1, with very few predictions near the decision boundary of 0.5. This indicates a highly confident and well-calibrated classifier. In contrast, the Audio SVM exhibits substantial overlap between the two classes in the 0.4 - 0.6 range, reflecting its comparatively lower accuracy of 72.7% and suggesting that acoustic features alone are insufficient to reliably distinguish deceptive from truthful speech in this dataset. The fused distribution, produced via weighted averaging, closely resembles the Visual NN distribution in shape, which is consistent with the higher weight assigned to the visual model during fusion. While audio scores introduce a slight spread toward the centre, they do not meaningfully sharpen the decision boundary. This visual analysis corroborates the quantitative finding that late fusion did not improve upon the best unimodal model, the visual signal is dominant and well-separated, leaving limited room for the audio modality to contribute additional discriminative information.

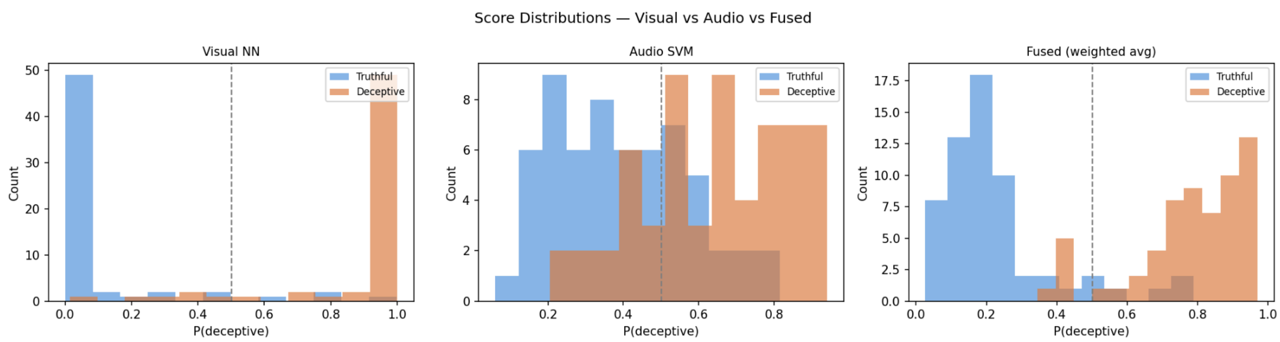


*Figure 2 : How Confident Is Each Model? - Score Distributions*

### 4.3 Diagnostic Analysis

Training and validation loss curves for the MLP show stable convergence without significant divergence between the two curves, indicating the regularization strategy (dropout, batch normalization, early stopping) is effective at the given data scale. The model does not show strong overfitting behavior, though the small dataset means high variance in per-fold estimates (standard deviations of ~3-5% across folds). The behavioral cross-training results indicate high bias in the cross-dataset condition, consistent with domain shift rather than model underfitting.

## 5. Safety, Security, and Ethics

### 5.1 Ethical Concerns

Representation bias: The Real-life Trial dataset contains 121 clips drawn from a narrow population of courtroom defendants. Vocal patterns, facial expressions, and body language vary substantially across gender, age, cultural background, and language. A model trained on this data will likely perform significantly worse for demographic groups not represented in training, producing systematically higher false positive rates for underrepresented populations.

Scientific validity: The scientific premise underlying automated deception detection, that lying produces consistent, detectable physiological and behavioral signals, remains heavily contested.

There is no universally accepted physiological marker of deception. Meta-analyses of polygraph validity consistently find effect sizes well below what would justify high-stakes decisions. Deploying a system built on a contested scientific foundation in consequential contexts is ethically problematic regardless of empirical performance on a benchmark.

Risk of serious harm: False positives in this domain carry severe consequences. A person wrongfully classified as deceptive in a legal, employment, or security screening context may face job loss, wrongful prosecution, or reputational damage. Given the model's best accuracy of 91.7% on a controlled dataset, approximately 1 in 11 predictions will be incorrect even under favorable conditions. At an organizational scale, this error rate produces a large absolute number of harmful outcomes.

Covert surveillance: Audio and video deception detection requires only a camera and microphone, enabling deployment without a subject's knowledge. This creates serious risks of mass surveillance, enabling governments or corporations to monitor individuals without consent or transparency.

### 5.2 Safety and Security Concerns

Adversarial manipulation: The audio SVM relies on directly measurable signals including pitch, speech rate, pause frequency, and MFCCs. A motivated actor aware that the system is operating can consciously modulate these features to evade detection. The model has no robustness against deliberate manipulation, a fundamental limitation for any security-critical deployment.

Systemic bias: Model errors are unlikely to be randomly distributed across the population. Systems trained on majority-group data tend to produce higher error rates for minority groups. In institutional settings, this means that the people most likely to be wrongly flagged are those already subject to systemic disadvantage, actively reinforcing inequality.

Deployment requirements: We conclude that this system, and deception detection systems generally at current capability levels, should not be used as standalone decision-making tools in any consequential setting. Any deployment would require: mandatory human review at every prediction, regular bias auditing across demographic groups, full explainability of individual decisions, explicit informed consent from all subjects, and clear legal frameworks governing use. The current state of the technology does not meet the bar required for production use in high-stakes domains.

## 6. Future Scope

- Scale dataset: Expanding beyond 121 clips to include diverse demographics, languages, and recording conditions is the single most impactful improvement available. The current dataset size severely limits generalization and statistical confidence.
- Incorporate transcript-based NLP features: The lexical content of deceptive statements (word choice, hedging language, narrative structure) is a well-studied deception signal. Adding a language model-based modality would make the system truly trimodal.
- Temporal modeling: Current features are summary statistics per clip. LSTMs or Transformer-based models operating over frame-level sequences could capture how deception signals evolve across a statement, potentially recovering information lost by aggregation.